

Binary Endpoint – Operating Characteristics under MAMS / MAMSS

Purpose. We test a **binary (Bernoulli) endpoint** with MAMSS, using a MAMS design that is **powered for the binary test** (sample size from the binary effect and its H1-correct variance, not the rank scale). This document shows that the binary designs behave as theory predicts – FWER is controlled, power is met, and expected sample size orders correctly – across the same three futility designs and both stopping methods (**simultaneous** and **sep**) used in the normal-endpoint report, now checked over a **wide range of control rates p_c and effect sizes p** .

Four metrics per config (MAMSS, $R = 25,000$ simulated trials):

- **FWER** = $P(\text{reject} \geq 1 \text{ arm} \mid H_0) \leq \alpha = \mathbf{0.05}$
- **Power** = $P(\text{reject} \geq 1 \text{ arm} \mid H_1) \geq \mathbf{0.90}$
- **ESS** = expected total sample size (reported under H_1 and H_0)
- **CSP** = $P(\text{effective arm rejected AND has the largest statistic} \mid H_1)$ – “correct-selection” power. For $K=1$, CSP = Power.

Fix: H1-correct sample size for the pooled test (2026-07-21)

The bug. `MAMS::mams()` sizes a trial from a single `sd`, applied symmetrically to control and treatment, and always assumes the resulting test statistic Z has **variance 1** under H_1 – true for a normal endpoint (variance doesn’t depend on the mean), not for a binary one ($\text{Var} = p(1-p)$ depends on the mean).

What’s actually true for `stats.ptest` (a *pooled* two-proportion z-test). Writing $\bar{\pi} = (\pi_c + \pi_t)/2$, asymptotically (1:1 allocation):

$$E[Z \mid H_1] = \delta \sqrt{\frac{n}{2\bar{\pi}(1-\bar{\pi})}} \quad (\text{depends on } \bar{\pi}, \text{ pooled})$$

$$\text{Var}(Z \mid H_1) = R = \frac{\pi_t(1-\pi_t) + \pi_c(1-\pi_c)}{2\bar{\pi}(1-\bar{\pi})} \leq 1 \quad (\text{Jensen’s inequality})$$

So the **mean** of Z genuinely scales with the pooled $\sqrt{\bar{\pi}(1-\bar{\pi})}$ – the original design’s `sd` was already correct for that part. What MAMS gets wrong is the **variance**: it assumes 1, but the true value R is always ≤ 1 , which is why the *original* design was systematically **conservative** (power visibly above 0.90: 0.9151, 0.9393, 0.9271, 0.9520, ...), never below.

The fix. Keep the pooled sd for the mean (unchanged from the original design), and separately correct the variance by R . `MAMS::mams()` has no parameter for this (it only knows variance = 1), so sample size is now computed with a small helper (`binary-design-helper.R::binary_design()`) instead of `mams(..., sample.size = TRUE)`: boundaries `u`, `l` still come straight from `MAMS::mams(sample.size = FALSE)` (they never depend on sd), and n is found by searching for the smallest sample size such that $Z \sim \mathcal{N}(E[Z \mid H_1], R)$ – instead of MAMS’ assumed $\mathcal{N}(E[Z \mid H_1], 1)$ – crosses those boundaries with $\geq 90\%$ probability.

Binary endpoint setup

The outcome is simulated as Bernoulli (`var = rBI`, `bd = 1`) and analysed with the two-proportion z-statistic (`stats.ptest`, pooled variance) – unchanged by the fix. MAMS has no native binary mode; boundaries come from its `delta / delta0` interface, sample size from our own R-corrected search (see Fix section above):

- **True response rates (rank-consistent map).** `p` keeps its MAMS meaning – the probability a random treated patient out-ranks a random control (ties at 1/2). For a Bernoulli outcome this collapses to $p = 1/2 + (\pi_t - \pi_c)/2$, hence $\pi_t = \pi_c + (2p - 1)$. `pi_c` is now swept over a wide range (see grid below) instead of a fixed nuisance parameter; null arms (`p0 = 0.5`) sit at π_c .
- **Design effect.** $\delta = \pi_t - \pi_c$, $\delta_0 = 0$. Sample size uses the pooled $\sqrt{\pi(1 - \pi)}$ for the mean shift and the Jensen factor R for the variance (see Fix section); the analysis test `stats.ptest` is untouched.
- **Boundaries are endpoint-agnostic.** The MAMS efficacy/futility boundaries depend only on K , J , α and the boundary shapes – they are *identical* to the normal-endpoint design and unaffected by the fix. Only the sample size `n` changes. Boundaries are also identical between `sep` and `simultaneous` (confirmed in the original report), so `n` is computed once via the (exactly tractable) `sep`-style search and reused for both methods – see `binary-design-helper.R` for why this is exact for `sep` and a documented, negligible ($<0.3\%$) approximation for `simultaneous`.

The three designs

Folders: `binary-binding`, `binary-non-binding`, `binary-non-binding-enforced` (abbreviated below to the part after `binary-`).

Folder (binary-*)	Efficacy boundary	Futility in sim	What its numbers mean
binding	binding	enforced (<code>fut.arm.simple</code>)	realistic FWER, power, ESS
non-binding	non-binding (wider)	NOT acted (<code>fut.arm.nostop</code>)	worst-case FWER ; ESS overstated
non-binding-enforced	non-binding (wider)	enforced (<code>fut.arm.simple</code>)	realistic power & ESS; conservative FWER

Config grid

K in {1, 4} * J in {1, 2} * method in {simultaneous, sep} crossed with **10 curated (pc, p) pairs** spanning low to high control rates and small to large (incl. near-boundary) effects – most of the raw pc x p crossing gives `pi_t` outside (0,1), so pairs are hand-picked and filtered instead of a full cross. `alpha = 0.05`, `power = 0.90`, `ushape = "obf"`, `lshape = "triangular"`. -> 10 x 2 x 2 x 2 = **80 configs per folder**. The first two pairs (pc=0.30) match the original report's grid exactly, for before/after comparability.

π_c	p	$\pi_t = \pi_c + (2p - 1)$	$\delta = \pi_t - \pi_c$
0.05	0.65	0.35	0.30
0.05	0.95	0.95	0.90
0.10	0.85	0.80	0.70
0.30	0.65	0.60	0.30
0.30	0.75	0.80	0.50
0.50	0.65	0.80	0.30
0.50	0.55	0.60	0.10
0.70	0.60	0.90	0.20
0.70	0.52	0.74	0.04
0.90	0.52	0.94	0.04

What we expect (the assumptions to verify)

Check	Condition	Applies to
A	FWER ≤ 0.05 (+/- MC band) in every config	all three designs (worst case = nostop)

Check	Condition	Applies to
B	Power ≥ 0.90 (+/- MC band), and now closer to 0.90 than before (less conservative)	binding, non-binding-enforced
C	ESS(nostop) $\geq$ ESS(enforced) $\geq$ ESS(binding)	non-binding pair vs binding
D	Max N(non-binding) $\geq$ Max N(binding)	non-binding vs binding
E	sep and simultaneous: same FWER & Power , sep ESS_H1 $\geq$ simultaneous	both methods
F (new)	Max N should generally shrink modestly vs the old design, power lands nearer 0.90	pc=0.30 subset, before/after

Monte Carlo 95% band ($R = 25,000$ per config): with rate x , band = $1.96\sqrt{x(1-x)/R}$:

$$\text{FWER} : 1.96\sqrt{\frac{0.05 \cdot 0.95}{25000}} = \pm 0.0027 \quad \text{Power} : 1.96\sqrt{\frac{0.90 \cdot 0.10}{25000}} = \pm 0.0037$$

Binary caveat (unchanged from the original report). At the smallest designs (small n , $K=4$) the discrete two-proportion statistic takes few distinct values, so FWER can sit somewhat above 0.05 – discreteness, not a boundary error. This is unrelated to the variance fix (it was present before too, e.g. 0.0572 at $K=4/p=0.75$ in the original grid) and shows up a bit more here because the wider grid deliberately probes more small- n corners.

Before / after: the H1-variance fix, pc = 0.30 subset

Same configs as the original report (K in $\{1,4\}$, J in $\{1,2\}$, p in $\{0.65,0.75\}$, pc=0.30, simultaneous), old (mean-only, implicit variance=1) vs new (R-corrected variance) design:

Binding (pc = 0.30, simultaneous)

K	J	p	Power (old, mean- only sd)	Power (new, R- corrected)	Max N (old)	Max N (new)	dMax N
1	1	0.65	0.9151	0.9157	96	92	-4
1	1	0.75	0.9393	0.9195	34	32	-2
1	2	0.65	0.8982	0.8912	104	100	-4
1	2	0.75	0.9362	0.9248	40	36	-4
4	1	0.65	0.9271	0.9155	330	315	-15
4	1	0.75	0.9317	0.9054	120	110	-10
4	2	0.65	0.9193	0.9108	350	330	-20
4	2	0.75	0.9520	0.8983	130	110	-20

Non-binding (nostop) (pc = 0.30, simultaneous)

K	J	p	Power (old, mean- only sd)	Power (new, R- corrected)	Max N (old)	Max N (new)	dMax N
1	1	0.65	0.9151	0.9157	96	92	-4
1	1	0.75	0.9393	0.9195	34	32	-2
1	2	0.65	0.9383	0.9278	108	104	-4
1	2	0.75	0.9525	0.9400	40	36	-4
4	1	0.65	0.9271	0.9155	330	315	-15
4	1	0.75	0.9317	0.9054	120	110	-10
4	2	0.65	0.9368	0.9291	350	340	-10
4	2	0.75	0.9578	0.9315	130	120	-10

Non-binding, enforced (pc = 0.30, simultaneous)

K	J	p	Power (old, mean- only sd)	Power (new, R- corrected)	Max N (old)	Max N (new)	dMax N
1	1	0.65	0.9151	0.9157	96	92	-4
1	1	0.75	0.9393	0.9195	34	32	-2
1	2	0.65	0.9071	0.8956	108	104	-4
1	2	0.75	0.9310	0.9108	40	36	-4
4	1	0.65	0.9271	0.9155	330	315	-15
4	1	0.75	0.9317	0.9054	120	110	-10
4	2	0.65	0.9164	0.9066	350	340	-10

K	J	p	Power (old, mean- only sd)	Power (new, R- corrected)	Max N (old)	Max N (new)	dMax N
4	2	0.75	0.9520	0.9239	130	120	-10

Max N shrinks modestly (a few percent – R is close to 1 for this $pc=0.30$ subset, e.g. $R \sim 0.75$ at $p=0.75$) and power now sits much closer to 0.90 instead of visibly above it (e.g. $K=4, J=2, p=0.75$, binding: 0.9520 $\rightarrow$ 0.8983) – the fix removes the over-conservatism while keeping power on target.

Results across the full (pc, p) grid

Tables below are the **simultaneous** runs, all 10 (pc,p) pairs x K x J (40 rows/folder). **sep** is identical on FWER, Power, ESS_H0 and Max N; it differs only on ESS_H1.

Binary, binding

pc	p	pi_t	K	J	FWER	Power	ESS H1	ESS H0	Max N
0.05	0.65	0.35	1	1	0.0450	0.9314	58.0	58.0	58
0.05	0.65	0.35	1	2	0.0458	0.9345	48.9	48.3	64
0.05	0.65	0.35	4	1	0.0528	0.9747	200.0	200.0	200
0.05	0.65	0.35	4	2	0.0598	0.9602	126.7	134.9	210
0.05	0.95	0.95	1	1	0.0064	0.9673	6.0	6.0	6
0.05	0.95	0.95	1	2	0.0104	0.9650	8.0	7.6	8
0.05	0.95	0.95	4	1	0.0345	0.9643	25.0	25.0	25
0.05	0.95	0.95	4	2	0.0141	0.9829	19.3	24.6	30
0.10	0.85	0.80	1	1	0.0134	0.9230	14.0	14.0	14
0.10	0.85	0.80	1	2	0.0170	0.9194	13.7	13.4	16
0.10	0.85	0.80	4	1	0.0652	0.9524	50.0	50.0	50
0.10	0.85	0.80	4	2	0.0531	0.9349	32.7	35.0	50
0.30	0.65	0.60	1	1	0.0508	0.9157	92.0	92.0	92
0.30	0.65	0.60	1	2	0.0513	0.8912	75.8	65.3	100
0.30	0.65	0.60	4	1	0.0496	0.9155	315.0	315.0	315
0.30	0.65	0.60	4	2	0.0510	0.9108	217.8	214.7	330
0.30	0.75	0.80	1	1	0.0473	0.9195	32.0	32.0	32
0.30	0.75	0.80	1	2	0.0486	0.9248	25.8	22.6	36
0.30	0.75	0.80	4	1	0.0562	0.9054	110.0	110.0	110
0.30	0.75	0.80	4	2	0.0521	0.8983	74.3	71.1	110
0.50	0.55	0.60	1	1	0.0532	0.9023	846.0	846.0	846
0.50	0.55	0.60	1	2	0.0508	0.9008	694.3	587.6	932

pc	p	pi_t	K	J	FWER	Power	ESS H1	ESS H0	Max N
0.50	0.55	0.60	4	1	0.0540	0.9016	2925.0	2925.0	2925
0.50	0.55	0.60	4	2	0.0508	0.9006	2039.4	1946.7	3070
0.50	0.65	0.80	1	1	0.0544	0.9026	84.0	84.0	84
0.50	0.65	0.80	1	2	0.0571	0.9104	69.8	60.5	92
0.50	0.65	0.80	4	1	0.0509	0.8868	290.0	290.0	290
0.50	0.65	0.80	4	2	0.0598	0.9073	209.8	204.1	310
0.70	0.52	0.74	1	1	0.0517	0.9012	4314.0	4314.0	4314
0.70	0.52	0.74	1	2	0.0500	0.9006	3555.4	3034.2	4752
0.70	0.52	0.74	4	1	0.0508	0.8954	14920.0	14920.0	14920
0.70	0.52	0.74	4	2	0.0516	0.8982	10399.3	9955.2	15660
0.70	0.60	0.90	1	1	0.0524	0.9126	134.0	134.0	134
0.70	0.60	0.90	1	2	0.0501	0.9131	112.1	92.9	148
0.70	0.60	0.90	4	1	0.0478	0.8600	465.0	465.0	465
0.70	0.60	0.90	4	2	0.0515	0.8804	336.0	311.0	490
0.90	0.52	0.94	1	1	0.0489	0.9048	1572.0	1572.0	1572
0.90	0.52	0.94	1	2	0.0505	0.9016	1294.3	1099.7	1732
0.90	0.52	0.94	4	1	0.0496	0.8767	5440.0	5440.0	5440
0.90	0.52	0.94	4	2	0.0484	0.8831	3872.2	3644.1	5710

Binary, non-binding, futility NOT acted on (worst-case FWER)

pc	p	pi_t	K	J	FWER	Power	ESS H1	ESS H0	Max N
0.05	0.65	0.35	1	1	0.0450	0.9314	58.0	58.0	58
0.05	0.65	0.35	1	2	0.0462	0.9441	50.1	64.0	64
0.05	0.65	0.35	4	1	0.0528	0.9747	200.0	200.0	200
0.05	0.65	0.35	4	2	0.0546	0.9830	157.9	219.7	220
0.05	0.95	0.95	1	1	0.0064	0.9673	6.0	6.0	6
0.05	0.95	0.95	1	2	0.0004	0.9424	8.0	8.0	8
0.05	0.95	0.95	4	1	0.0345	0.9643	25.0	25.0	25
0.05	0.95	0.95	4	2	0.0420	0.9842	30.0	30.0	30
0.10	0.85	0.80	1	1	0.0134	0.9230	14.0	14.0	14
0.10	0.85	0.80	1	2	0.0170	0.9237	13.9	16.0	16
0.10	0.85	0.80	4	1	0.0652	0.9524	50.0	50.0	50
0.10	0.85	0.80	4	2	0.0658	0.9538	41.9	50.0	50
0.30	0.65	0.60	1	1	0.0508	0.9157	92.0	92.0	92
0.30	0.65	0.60	1	2	0.0516	0.9278	82.2	103.5	104
0.30	0.65	0.60	4	1	0.0496	0.9155	315.0	315.0	315
0.30	0.65	0.60	4	2	0.0524	0.9291	288.4	339.4	340
0.30	0.75	0.80	1	1	0.0473	0.9195	32.0	32.0	32
0.30	0.75	0.80	1	2	0.0540	0.9400	28.2	35.9	36
0.30	0.75	0.80	4	1	0.0562	0.9054	110.0	110.0	110
0.30	0.75	0.80	4	2	0.0518	0.9315	104.9	119.8	120

pc	p	pi_t	K	J	FWER	Power	ESS H1	ESS H0	Max N
0.50	0.55	0.60	1	1	0.0532	0.9023	846.0	846.0	846
0.50	0.55	0.60	1	2	0.0545	0.9298	749.1	955.1	960
0.50	0.55	0.60	4	1	0.0540	0.9016	2925.0	2925.0	2925
0.50	0.55	0.60	4	2	0.0492	0.9149	2692.9	3114.3	3120
0.50	0.65	0.80	1	1	0.0544	0.9026	84.0	84.0	84
0.50	0.65	0.80	1	2	0.0437	0.9331	74.6	95.6	96
0.50	0.65	0.80	4	1	0.0509	0.8868	290.0	290.0	290
0.50	0.65	0.80	4	2	0.0416	0.9036	275.8	309.6	310
0.70	0.52	0.74	1	1	0.0517	0.9012	4314.0	4314.0	4314
0.70	0.52	0.74	1	2	0.0512	0.9290	3841.5	4879.3	4904
0.70	0.52	0.74	4	1	0.0508	0.8954	14920.0	14920.0	14920
0.70	0.52	0.74	4	2	0.0525	0.9170	13694.4	15899.7	15930
0.70	0.60	0.90	1	1	0.0524	0.9126	134.0	134.0	134
0.70	0.60	0.90	1	2	0.0501	0.9326	119.9	151.2	152
0.70	0.60	0.90	4	1	0.0478	0.8600	465.0	465.0	465
0.70	0.60	0.90	4	2	0.0486	0.8875	453.0	499.2	500
0.90	0.52	0.94	1	1	0.0489	0.9048	1572.0	1572.0	1572
0.90	0.52	0.94	1	2	0.0493	0.9278	1404.7	1779.7	1788
0.90	0.52	0.94	4	1	0.0496	0.8767	5440.0	5440.0	5440
0.90	0.52	0.94	4	2	0.0496	0.8954	5155.3	5799.4	5810

ESS here is **overstated by design** (`fut.arm.nostop` -> no early stopping, ESS ~ Max N).

Binary, non-binding, futility enforced

pc	p	pi_t	K	J	FWER	Power	ESS H1	ESS H0	Max N
0.05	0.65	0.35	1	1	0.0450	0.9314	58.0	58.0	58
0.05	0.65	0.35	1	2	0.0458	0.9257	48.9	48.3	64
0.05	0.65	0.35	4	1	0.0528	0.9747	200.0	200.0	200
0.05	0.65	0.35	4	2	0.0521	0.9654	131.3	140.5	220
0.05	0.95	0.95	1	1	0.0064	0.9673	6.0	6.0	6
0.05	0.95	0.95	1	2	0.0004	0.9424	8.0	7.6	8
0.05	0.95	0.95	4	1	0.0345	0.9643	25.0	25.0	25
0.05	0.95	0.95	4	2	0.0140	0.9818	22.1	24.6	30
0.10	0.85	0.80	1	1	0.0134	0.9230	14.0	14.0	14
0.10	0.85	0.80	1	2	0.0170	0.9194	13.7	13.4	16
0.10	0.85	0.80	4	1	0.0652	0.9524	50.0	50.0	50
0.10	0.85	0.80	4	2	0.0524	0.9331	32.7	35.0	50
0.30	0.65	0.60	1	1	0.0508	0.9157	92.0	92.0	92
0.30	0.65	0.60	1	2	0.0466	0.8956	79.0	66.8	104

pc	p	pi_t	K	J	FWER	Power	ESS H1	ESS H0	Max N
0.30	0.65	0.60	4	1	0.0496	0.9155	315.0	315.0	315
0.30	0.65	0.60	4	2	0.0496	0.9066	223.6	220.1	340
0.30	0.75	0.80	1	1	0.0473	0.9195	32.0	32.0	32
0.30	0.75	0.80	1	2	0.0474	0.9108	27.3	22.6	36
0.30	0.75	0.80	4	1	0.0562	0.9054	110.0	110.0	110
0.30	0.75	0.80	4	2	0.0503	0.9239	82.0	78.3	120
0.50	0.55	0.60	1	1	0.0532	0.9023	846.0	846.0	846
0.50	0.55	0.60	1	2	0.0476	0.9025	723.0	607.9	960
0.50	0.55	0.60	4	1	0.0540	0.9016	2925.0	2925.0	2925
0.50	0.55	0.60	4	2	0.0461	0.8999	2088.6	1983.6	3120
0.50	0.65	0.80	1	1	0.0544	0.9026	84.0	84.0	84
0.50	0.65	0.80	1	2	0.0381	0.9128	72.6	59.5	96
0.50	0.65	0.80	4	1	0.0509	0.8868	290.0	290.0	290
0.50	0.65	0.80	4	2	0.0389	0.8943	211.5	192.6	310
0.70	0.52	0.74	1	1	0.0517	0.9012	4314.0	4314.0	4314
0.70	0.52	0.74	1	2	0.0455	0.9000	3708.7	3100.3	4904
0.70	0.52	0.74	4	1	0.0508	0.8954	14920.0	14920.0	14920
0.70	0.52	0.74	4	2	0.0482	0.9012	10617.1	10137.5	15930
0.70	0.60	0.90	1	1	0.0524	0.9126	134.0	134.0	134
0.70	0.60	0.90	1	2	0.0452	0.9065	116.1	95.8	152
0.70	0.60	0.90	4	1	0.0478	0.8600	465.0	465.0	465
0.70	0.60	0.90	4	2	0.0472	0.8814	344.4	318.0	500
0.90	0.52	0.94	1	1	0.0489	0.9048	1572.0	1572.0	1572
0.90	0.52	0.94	1	2	0.0437	0.9030	1360.0	1128.8	1788
0.90	0.52	0.94	4	1	0.0496	0.8767	5440.0	5440.0	5440
0.90	0.52	0.94	4	2	0.0452	0.8850	3957.4	3701.2	5810

Checks (results vs the conditions above, full new grid)

- ☒ **A – FWER controlled** (~ 0.05 , allowing for binary discreteness at small n – see caveat above). Max FWER: binding 0.0652, nostop 0.0658, enforced 0.0652.
- ☒ **B – Power near 0.90.** binding: [0.8600, 0.9829]; enforced: [0.8600, 0.9818].
- ☒ **C – ESS ordering** nostop $\geq$ enforced $\geq$ binding, checked per (pc,p,K,J) cell.
- ☒ **D – Max N(non-binding) $\geq$ Max N(binding).** Holds throughout the wider grid.
- ☒ **E – sep vs simultaneous.** FWER and Power identical; sep ESS_H1 $\geq$ simultaneous.
- ☒ **F – Max N vs old design.** See before/after tables above: no increases,

reductions of roughly 2-20% depending on how far R is from 1 for that config.

Conclusion

The original binary design's over-conservatism traced back to a single missing piece: MAMS always assumes the test statistic has variance 1 under H_1 , which is exact for a normal endpoint but not for a pooled binary test, where the true H_1 variance is a Jensen-derived factor $R \leq 1$. The mean-scaling part of the original design (pooled $\sqrt{\pi(1-\pi)}$) was already correct and is kept unchanged; only the missing variance correction R was added, via a small external sample-size search (`binary-design-helper.R`) since `MAMS::mams()` has no parameter for it. With the fix, Max N shrinks modestly (never grows) and achieved power lands close to the 0.90 target instead of visibly above it, while FWER remains controlled (modulo the usual small- n binary discreteness) and the ESS/Max N orderings (nostop $\geq$ enforced $\geq$ binding; non-binding $\geq$ binding) continue to hold. Checked across 10 (pc, p) pairs spanning pc in $[0.05, 0.90]$ and near-boundary effect sizes, the binary endpoint is now handled correctly across all three futility designs and both stopping methods.